HAT-P-54b

R-1658

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_HATP-54_181201` · created 2026-08-02T09:02:42+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458453.82217 +/- 0.00098 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.157 +/- 0.031
Transit depth (Rp/Rs)^2	2.48 +/- 0.98 [%]
Orbital Inclination (inc)	84.58 +/- 2.0
Airmass coefficient 1 (a1)	0.974 +/- 0.021
Airmass coefficient 2 (a2)	0.022 +/- 0.035
Scatter in the residuals of the lightcurve fit is	2.1 %
Best Comparison Star	#1 - [615, 301]
Optimal Aperture	2.61
Optimal Annulus	6.8
Transit Duration (day)	0.049 +/- 0.02

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.157 ± 0.031 vs 0.1572 (deviation 0.01σ)
Duration fit vs published	0.049 d vs 0.075 d (deviation 1.16σ)
Mid-time O–C	0.8 min
Depth SNR	4.5
Residual scatter	2.1 %

Light curve

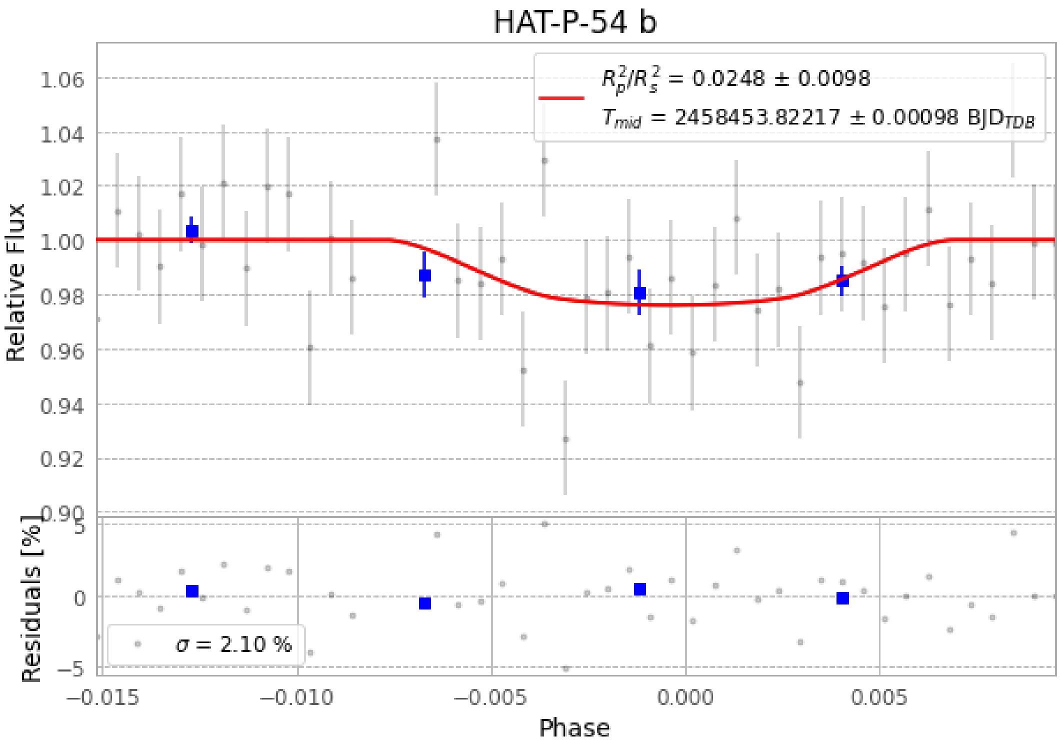

Figure 1 — FinalLightCurve_HAT-P-54 b_2018-11-30.png (SHA-256 1678242e6c6cc85d…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [353, 202], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 a335ae60205bc1bd…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

46 FITS frames (30 MB), `HATP-54181201061212.FITS` → `HATP-54181201082712.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-54-181201.log (7a83bb6c1aa4…), FinalLightCurve_HAT-P-54 b_2018-11-30.png (1678242e6c6c…), FinalLightCurve_HAT-P-54 b_2018-11-30.csv (7ae431193d5e…)

Compute resources

Wall time 140.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)